



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Course: c++ programming DSA0108

1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a **Consultation Charge System** for hospital management.

Task:

- Use **function overloading** to calculate charges for **general consultation** and **specialist consultation**.
- Use **call by reference** to modify the final bill amount.
- Use **default arguments** to add a standard service charge if needed.
- Use **function prototyping** for all calculation functions.
- Display the **patient type** and **total consultation charge**. (BL3)

```
1 #include<iostream>
2 using namespace std;
3 float charge(float g,float s=100);
4 float charge(float g,int sp,float s=100);
5 void bill(float &b);
6 float charge(float g,float s){return g+s;}
7 float charge(float g,int sp,float s){return g+sp+s;}
8 void bill(float &b){b*=1.05;}
9 int main(){
10 int ch,sp;
11 float fee,b;
12 cin>>ch>>fee;
13 if(ch==1){
14     b=charge(fee);
15     bill(b);
16     cout<<"specialist\n:"<<b;}
17 else{
18     cin>>sp;
19     b=charge(fee,sp);
```

1
500

```

13  if(ch==1){
14      b=charge(fee);
15      bill(b);
16      cout<<"specialist\n:"<<b;}}
17  else{
18      cin>>sp;
19      b=charge(fee,sp);
20      bill(b);
21      cout<<"specialist\n:"<<b;}}

```

OUTPUT

```

specialist
:630

```

2. Design a C++ program for a Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

- Create a class **Member** with the attributes: **Member ID, Name, Plan Type, and Fee.**
- Implement member functions to **assign plans, calculate fees, and display membership details.**
- Use an **array of objects** to manage multiple members.
- Add **validation** for valid membership plans.
- Include a **static member** to track the total number of active members.
- Evaluate how your design supports **easy modification of pricing plans.**

The screenshot shows a C++ programming IDE with a 'PROGRAM EDITOR' window and an 'INPUT' window. The program editor contains the following code:

```

60
61 void display()
62 {
63     cout << "\n----- Member Details -----" << endl;
64     cout << "Member ID : " << memberID << endl;
65     cout << "Name      : " << name << endl;
66     cout << "Plan Type : " << planType << endl;
67     cout << "Fee       : Rs. " << fee << endl;
68 }
69 };
70
71 int Member::activeMembers = 0;
72
73 int main()
74 {
75     int n;

```

The input window contains the following text:

```

2
1
100
madhulika
premium
2
102
priya
gold

```

```
72 int MemberID, ReceiveNumber, S;
73
74 int main()
75 {
76     int n;
77
78     //===== MEMBERSHIP DETAILS =====
79
80     //---- Member Details ----
81     Member ID : 1
82     Name      : madhulika
83     Plan Type  : premium
84     Fee        : Rs. 2000
85     Membership Fee: Rs. 2000
86
87     //---- Member Details ----
88 }
```

OUTPUT

```
===== MEMBERSHIP DETAILS =====
---- Member Details ----
Member ID : 1
Name      : madhulika
Plan Type  : premium
Fee        : Rs. 2000
Membership Fee: Rs. 2000
---- Member Details ----
```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Task:

- Use **function prototyping** to declare EMI-related functions.
- Use **default arguments** for the default interest rate where applicable.
- Apply **call by reference** to update the payable amount after subsidy.
- Use **inline functions** for simple interest calculations.
- Display the **loan amount, interest, EMI, subsidy (if any), payable amount, and repayment details** clearly.



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C++ PROGRAMMING

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QUESTIONS 1 / 9 completed

GCD-Fun.

GCD-FUN.

Write a function to determine the GCD (greatest common divisor) of two given integers.

PROGRAM EDITOR

C++

Run Save

```
1 #include<iostream>
2 using namespace std;
3 inline float interest(float p,float r=10,float t=1);
4 void subsidy(float &amt);
5 inline float interest(float p,float r,float t)
6 {
7     return(p*t*r)/100;
8 }
9 void subsidy(float &amt)
10 {
11     amt=amt-1000;
12 }
13 int main()
14 {
15     float loan,i,total;
16     cout<<"enter loan amount:"<<endl;
17     cin>>loan;
```

INPUT

```
10000
10000
1000
10000
```

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C++ PROGRAMMING

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4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

- Create a class Parcel with appropriate private data members such as parcel type, weight, distance, and packaging charge.
- Use public member functions to read parcel details, calculate the delivery charge, and display the parcel information.
- Use function overloading to calculate charges for Normal Delivery and Express Delivery.
- Use default arguments to assign a default packaging charge when it is not specified by the user.
- Use an inline function to calculate the base charge based on parcel weight and delivery distance.
- Maintain the total number of parcels processed using a static data member and display it through a static member function.

```

75
76   for (int i = 0; i < n; i++) {
77       cout << "\nEnter Details of Parcel " << i + 1 << endl;
78       p[i].getData();
79
80       if (p[i].getType() == "Express" || p[i].getType() == "express")
81           p[i].calculateCharge(1);
82       else
83           p[i].calculateCharge();
84   }
85
86   cout << "\n===== PARCEL DETAILS =====>

```

```

92   return 0;
93 }
94

```

OUTPUT

```

Enter Number of Parcels:
Enter Details of Parcel 1
Enter Parcel Type (Normal/Express): Enter Weight (kg): Enter
Distance (km):
Enter Details of Parcel 2
Enter Parcel Type (Normal/Express): Enter Weight (kg): Enter
Distance (km):
===== PARCEL DETAILS =====

```

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a **Water Billing System**.

- Create a class **Consumer** with appropriate private data members such as **consumer ID**, **consumer type**, **water consumption (in liters)**, and **bill amount**.
- Use public member functions to **input consumer details**, **calculate the water bill**, and **display the billing information**.
- Implement **function overloading** to calculate water bills for **Domestic** and **Commercial** consumers using different tariff rates.
- Use **call by reference** to update the bill amount after applying a government subsidy.
- Use **default arguments** to include a service tax, where a default tax rate is applied unless specified.
- Declare all functions using **function prototyping** before their definitions.

g. Maintain the total number of consumers processed using a **static data member** and display it through a **static member function**.

h. Create an **array of objects** to store and process the details of multiple consumers, and display the **consumer type, water consumption, and subsidy-adjusted bill amount**.

```
47     else
48         calculateBill();
49
50     applySubsidy(bill, subsidy);
51 }
52
53 void display() {
54     cout << "\nConsumer ID : " << id;
55     cout << "\nType      : " << type;
56     cout << "\nWater Used : " << water << " Liters";
57     cout << "\nBill Amount : Rs. " << bill << endl;
58 }
59
60 static void showTotal() {
61     cout << "\nTotal Consumers Processed: " << total << endl;
62 }
63 };
64
65 int Consumer::total = 0;
```

```
2
101
domestic
500
50
101
commercial
800
100
```

```
94
95
```

OUTPUT

```
Consumer ID : 101
Type      : domestic
Water Used : 500 Liters
Bill Amount : Rs. 1000

Consumer ID : 101
Type      : commercial
Water Used : 800 Liters
Bill Amount : Rs. 3940
```